Activity 10— Food Waste Monitoring System

1. Problem Statement

Develop a Python program to monitor food waste by recording the quantity of food prepared and the quantity wasted. Calculate the percentage of food wasted and display a message based on the result.

2. Objectives

- To record food preparation and waste quantities.
- To calculate the amount of food consumed or used.
- To calculate the food waste percentage.
- To understand Python variables, arithmetic operators and conditional statements.

3. Assumptions

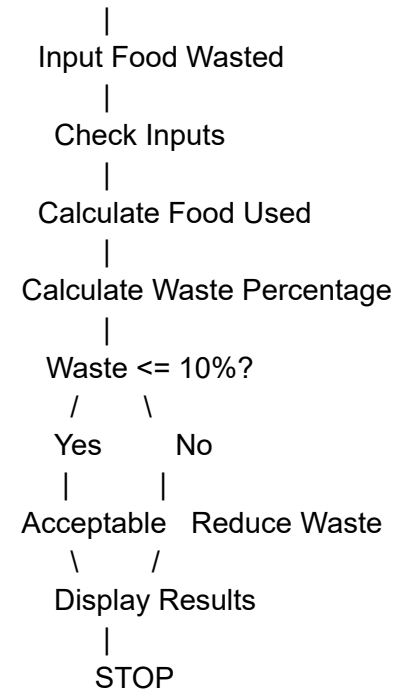
- Food prepared is measured in kilograms.
- Food wasted is measured in kilograms.
- Food prepared must be greater than zero.
- Food wasted cannot be negative or greater than the quantity prepared.
- A waste percentage of 10% or less is considered acceptable for this example.

4. Algorithm

1. Start.
2. Enter the quantity of food prepared.
3. Enter the quantity of food wasted.
4. Check whether the inputs are valid.
5. Calculate the food used by subtracting wasted food from prepared food.
6. Calculate the waste percentage.
7. Display the food used and waste percentage.
8. Display a message based on the waste percentage.
9. Stop.

5. Flowchart





6. Pseudocode

```
START
INPUT prepared
INPUT wasted

IF prepared <= 0 OR wasted < 0 OR wasted > prepared
   PRINT "Invalid input"
ELSE
   used = prepared - wasted
   percentage = (wasted / prepared) * 100

   PRINT used
   PRINT percentage

   IF percentage <= 10
      PRINT "Food waste is acceptable"
   ELSE
      PRINT "Try to reduce food waste"
   ENDIF
ENDIF

STOP
```

7. Python Source Code

File name: `activity10.py`

```
prepared = float(input("Enter food prepared (kg): "))
wasted = float(input("Enter food wasted (kg): "))

if prepared <= 0 or wasted < 0 or wasted > prepared:
    print("Invalid input")

else:
    used = prepared - wasted
    percentage = (wasted / prepared) * 100

    print("Food Used:", round(used, 2), "kg")
    print("Food Waste:", round(wasted, 2), "kg")
    print("Waste Percentage:", round(percentage, 2), "%")

    if percentage <= 10:
        print("Food waste is acceptable")
    else:
        print("Try to reduce food waste")
```

8. Test Cases

Food Prepared	Food Wasted	Expected Result
100 kg	5 kg	5% waste; acceptable
100 kg	20 kg	20% waste; try to reduce waste
50 kg	5 kg	10% waste; acceptable
100 kg	120 kg	Invalid input
0 kg	0 kg	Invalid input

9. Output Screenshot and Conclusion

Sample Output:

```
Enter food prepared (kg): 100
Enter food wasted (kg): 20
Food Used: 80.0 kg
Food Waste: 20.0 kg
Waste Percentage: 20.0 %
Try to reduce food waste
```

Output Screenshot: Run `activity10.py` in Python IDLE or VS Code. Capture the inputs and output, and save the screenshot as `activity10_output.png`.

Conclusion: This program calculates food waste and its percentage using Python arithmetic and conditional statements. It helps monitor waste and encourages better food management.

INPUT:

```
prepared = float(input("Enter food prepared (kg): "))
wasted = float(input("Enter food wasted (kg): "))

if prepared <= 0 or wasted < 0 or wasted > prepared:
    print("Invalid input")

else:
    used = prepared - wasted
    percentage = (wasted / prepared) * 100

    print("Food Used:", round(used, 2), "kg")
    print("Food Waste:", round(wasted, 2), "kg")
    print("Waste Percentage:", round(percentage, 2), "%")

    if percentage <= 10:
        print("Food waste is acceptable")
    else:
        print("Try to reduce food waste")
```

OUTPUT:

```
Enter food prepared (kg): 100
Enter food wasted (kg): 5
Food Used: 95.0 kg
Food Waste: 5.0 kg
Waste Percentage: 5.0 %
Food waste is acceptable

====================================
Enter food prepared (kg): 100
Enter food wasted (kg): 20
Food Used: 80.0 kg
Food Waste: 20.0 kg
Waste Percentage: 20.0 %
Try to reduce food waste

====================================
Enter food prepared (kg): 100
Enter food wasted (kg): 120
Invalid input

====================================
Enter food prepared (kg): 80
Enter food wasted (kg): 5
Food Used: 75.0 kg
Food Waste: 5.0 kg
Waste Percentage: 6.25 %
Food waste is acceptable

====================================
Enter food prepared (kg): 0
Enter food wasted (kg): 0
Invalid input
```